

# **DIMINUTIVES IN SPANISH: CYCLICITY, ALLOMORPHY, PROSODY**

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This short paper assesses the data on Peninsular Spanish diminutives against an articulated theory of nominal classes in Spanish within Stratal Phonology (Bermúdez-Otero 2013). As opposed to many previous monostratal approaches (see Smith 2011:§2), the analysis developed in this short paper upholds a high standard of modularity. First, no morphological (e.g., nominal class) or syntactic property (e.g., DIM), is named in the statement of the phonological constraints, except for ‘stem’ in alignment constraints (Indirect Reference Hypothesis, Bermúdez-Otero 2012:77). Secondly, the high standard of modularity is also grounded in cyclicity: morphosyntactic structure determines the cyclic domains in which allomorph selection is resolved by controlling the size and stratal affiliation of domains for phonological computation.

**Keywords:** Cyclicity, Allomorphy, Prosody, Diminutive, Spanish

## MOTIVATION

Providing a comprehensive account of diminutive formation in Peninsular Spanish is challenging because Spanish diminutives exhibit (i) cyclic misapplication, induced by morphosyntactic structure, namely overapplication in the selection of diphthongal stems (e.g., *piedrecita* ‘stone.DIM’, not \**pedrecita*); and (ii) an intricate pattern of allomorph selection of the diminutive suffix (*-it-o/a* or *-cit-o/a*).<sup>1</sup>

I will show that allomorph selection of the diminutive suffix in Peninsular Spanish is conditioned by multiple interacting factors: (i) morphology–prosody alignment, (ii) faithfulness to the underlying prosodic specification of the *-cit-o/a* allomorph, (iii) a ban on pairs of piece-final vowels sharing specific phonological features, and (iv) a ban on extended lapses.

The main goal of this short paper is to assess the data on Peninsular Spanish diminutives against an articulated theory of nominal classes in Spanish (Bermúdez-Otero 2013) within Stratal Phonology (Bermúdez-Otero 2011, 2018, 2025; Bermúdez-Otero & Torres-Tamarit 2026; Kiparsky 2000, 2015; Trommer 2025).

I claim that (i) overapplication in the selection of the diphthongal stem in diminutives is accounted for straightforwardly by level selection in Stratal Phonology (as in Bermúdez-Otero 2013), and (ii) a nuanced understanding of nominal classes in Spanish better captures the pattern of allomorph selection of the diminutive suffix and especially the distribution of [e] preceding the *-cit-o/a* allomorph, which is either a theme vowel or an epenthetic vowel.

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## 1 NOMINAL CLASSES IN SPANISH

I assume the classification by Bermúdez-Otero (2013) of nominal elements in Spanish into three core classes (*o*-, *a*-, and *e*-stems) and a peripheral athematic class, as illustrated in (1). In both *o*- and *a*-stems, the theme vowel appears in both the singular and the plural. The *e*-stem class is subdivided into two subclasses: ordinary *e*-stems, in which the theme vowel *-e* appears only in the plural—or in the singular when phonotactic constraints require it—and *e*-only stems, in which the theme vowel *-e* is present in both the singular and the plural. Finally, the athematic class is characterized by the absence of a theme vowel in both the singular and the plural.

(1) Nominal classes in Spanish (Bermúdez-Otero 2013:10)

| Class     |          | Theme   | SG        | PL          | Gloss       | Gender |
|-----------|----------|---------|-----------|-------------|-------------|--------|
| o-stem    |          | /-o/    | [lí-o]    | [lí-o-s]    | ‘muddle’    | M      |
|           |          |         | [mán-o]   | [mán-o-s]   | ‘hand’      | F      |
| a-stem    |          | /-a/    | [dí-a]    | [dí-a-s]    | ‘day’       | M      |
|           |          |         | [kán-a]   | [kán-a-s]   | ‘grey hair’ | F      |
| e-stem    | ordinary | /-e, Ø/ | [lápiθ-Ø] | [lápiθ-e-s] | ‘pencil’    | M      |
|           | e-stem   |         | [lúθ-Ø]   | [lúθ-e-s]   | ‘light’     | F      |
|           |          |         | [páðr-e]  | [páðr-e-s]  | ‘father’    | M      |
|           |          |         | [máðr-e]  | [máðr-e-s]  | ‘mather’    | F      |
|           | e-only   | /-e/    | [krúθ-e]  | [krúθ-e-s]  | ‘crossing’  | M      |
|           | stem     |         | [éliθ-e]  | [éliθ-e-s]  | ‘propeller’ | F      |
| athematic | stem     | /-Ø/    | [roβót-Ø] | [roβót-Ø-s] | ‘robot’     | M      |
|           |          |         | [tríβu-Ø] | [tríβu-Ø-s] | ‘tribe’     | F      |

I will show how this classification of nominal elements in Spanish sheds light on the distribution of the diminutive allomorph and on the correct characterization of [e] pre-

ceding the *-cit-o/a* allomorph.

## 2 DATA AND DESCRIPTIVE GENERALIZATIONS

*O-* and *a-*stems take *-it-o/a* regardless of syllable count, except in disyllabic bases with a glide, which select *-cit-o/a* and insert an [e] vowel between the root and the diminutive suffix—an element that cannot be interpreted as a theme vowel because these stems belong to the *o-* and *a-*stem classes, but rather as an epenthetic vowel ((2)a,b).

Ordinary *e-*stems take the null theme  $-\emptyset$  and select *-cit-o/a*, except when the singular base is monosyllabic, in which case the theme *-e* surfaces. When the singular base already ends in the theme vowel *-e*—a strategy used when certain root-final consonants are disfavoured word-finally—the theme likewise appears in the diminutive formed with *-cit-o/a*, unless the singular base exceeds two syllables; in that case, the theme *-e* does not surface and *-it-o/a* is selected ((2)c).

*E-*only stems retain the theme vowel *-e* and select *-cit-o/a*, except when the singular base exceeds two syllables; in that case, the theme *-e* does not surface and *-it-o/a* is selected ((2)d).

Athematic stems take *-cit-o/a*, except when they end in /s/ or in a consonant that is illicit in preconsonantal position word-medially; such stems instead select *-it-o/a* ((2)e). The data presented in (2) come from Colina (2003), Smith (2011), *NGLE* (2009-2011):§9.4, §9.5, and my own grammatical judgements.

(2) Peninsular Spanish diminutives

a. o-stem

|      | Base              | Diminutive                    |              |
|------|-------------------|-------------------------------|--------------|
| i.   | <i>libr-o</i>     | <i>libr-<u>it-o</u></i>       | ‘book.M’     |
|      | <i>zapat-o</i>    | <i>zapat-<u>it-o</u></i>      | ‘shoe.M’     |
| ii.  | <i>v[j]ent-o</i>  | <i>v[j]ent-<u>e-cit-o</u></i> | ‘wind.M’     |
|      | <i>sab[j]-o</i>   | <i>sab[j]-<u>e-cit-o</u></i>  | ‘wise man.M’ |
| iii. | <i>armar[j]-o</i> | <i>armar-<u>it-o</u></i>      | ‘wardrobe.M’ |

b. a-stem

|      | Base              | Diminutive                    |              |
|------|-------------------|-------------------------------|--------------|
| i.   | <i>cas-a</i>      | <i>cas-<u>it-a</u></i>        | ‘house.F’    |
|      | <i>pregunt-a</i>  | <i>pregunt-<u>it-a</u></i>    | ‘question.F’ |
| ii.  | <i>p[j]edr-a</i>  | <i>p[j]edr-<u>e-cit-a</u></i> | ‘stone.F’    |
|      | <i>mom[j]-a</i>   | <i>mom[j]-<u>e-cit-a</u></i>  | ‘mummy.F’    |
| iii. | <i>igles[j]-a</i> | <i>igles-<u>it-a</u></i>      | ‘church.F’   |

c. ordinary e-stem

|      | Base              | Diminutive                   |               |
|------|-------------------|------------------------------|---------------|
| i.   | <i>canción</i>    | <i>cancion-<u>cit-a</u></i>  | ‘song.F’      |
|      | <i>escritor</i>   | <i>escritor-<u>cit-o</u></i> | ‘writer.M’    |
| ii.  | <i>flor</i>       | <i>flor-<u>e-cit-a</u></i>   | ‘flower.F’    |
|      | <i>pan</i>        | <i>pan-<u>e-cit-o</u></i>    | ‘bread.M’     |
| iii. | <i>madr-e</i>     | <i>madr-<u>e-cit-a</u></i>   | ‘mother.F’    |
|      | <i>padr-e</i>     | <i>padr-<u>e-cit-o</u></i>   | ‘father.M’    |
| iv.  | <i>estuch-e</i>   | <i>estuch-<u>it-o</u></i>    | ‘case.M’      |
|      | <i>chocolat-e</i> | <i>chocolat-<u>it-o</u></i>  | ‘chocolate.M’ |

d. *e*-only stem

|     | Base             | Diminutive          |               |
|-----|------------------|---------------------|---------------|
| i.  | <i>fas-e</i>     | <i>fas-e-cit-a</i>  | ‘phase.F’     |
|     | <i>pas-e</i>     | <i>pas-e-cit-o</i>  | ‘pass.M’      |
| ii. | <i>pirámid-e</i> | <i>piramid-it-a</i> | ‘pyramid.F’   |
|     | <i>envas-e</i>   | <i>envas-it-o</i>   | ‘container.M’ |

e. athematic stem

|     | Base              | Diminutive           |               |
|-----|-------------------|----------------------|---------------|
| i.  | <i>café</i>       | <i>cafe-cit-o</i>    | ‘coffee.M’    |
|     | <i>sofá</i>       | <i>sofa-cit-o</i>    | ‘sofa.M’      |
|     | <i>yoyó</i>       | <i>yoyo-cit-o</i>    | ‘yoyo.M’      |
|     | <i>taxi</i>       | <i>taxi-cit-o</i>    | ‘taxi.M’      |
|     | <i>tribu</i>      | <i>tribu-cit-a</i>   | ‘tribe.F’     |
|     | <i>caqui</i>      | <i>caqui-cit-o</i>   | ‘persimmon.M’ |
|     | <i>haiku</i>      | <i>haiku-cit-o</i>   | ‘haiku.M’     |
|     | <i>sticker</i>    | <i>sticker-cit-o</i> | ‘sticker.M’   |
|     | <i>fan</i>        | <i>fan-cit-o</i>     | ‘fan.M’       |
|     | <i>bol</i>        | <i>bol-cit-o</i>     | ‘bowl.M’      |
| ii. | <i>clip</i>       | <i>clip-it-o</i>     | ‘clip.M’      |
|     | <i>robot</i>      | <i>robot-it-o</i>    | ‘robot.M’     |
|     | <i>anorak</i>     | <i>anorak-it-o</i>   | ‘anorak.M’    |
|     | <i>caries</i>     | <i>caries-it-a</i>   | ‘caries.F’    |
|     | <i>brindis</i>    | <i>brindis-it-o</i>  | ‘toast.M’     |
|     | <i>virus</i>      | <i>virus-it-o</i>    | ‘virus.M’     |
|     | <i>análisi-s</i>  | <i>analís-it-o</i>   | ‘analysis.M’  |
|     | <i>Sócrates-s</i> | <i>socrat-it-o</i>   | ‘Socrates.M’  |

In (3) the main patterns are summarized. In the present analysis, diminutives derived

from athematic stems are not considered in detail; further research producing first-hand data is needed before dealing with these cases.

(3) Summary chart

| <i>Class</i>             | <i>Diminutive allomorph</i> | <i>Example</i>                                  | <i>Base</i>           |
|--------------------------|-----------------------------|-------------------------------------------------|-----------------------|
| a. <i>o-/a-stem</i>      | -it-o/a                     | zapat- <u>it</u> -o<br>pregunt- <u>it</u> -a    | zapat-o<br>pregunt-a  |
| base                     |                             |                                                 |                       |
| with [j]                 | -e-cit-o/a                  | vient- <u>e</u> -cit-o<br>momi- <u>e</u> -cit-a | v[j]ent-o<br>mom[j]-a |
| > 2 $\sigma$             | -it-o/a                     | igles- <u>it</u> -a                             | igles[j]-a            |
| b. <i>e-stem</i>         | -cit-o/a                    | escritor- <u>cit</u> -o<br>fas- <u>e</u> -cit-a | escritor<br>fas-e     |
| monosyllabic             |                             |                                                 |                       |
| base                     | e-cit-o/a                   | flor- <u>e</u> -cit-a                           | flor                  |
| > 2 $\sigma$             | -it-o/a                     | piramid- <u>it</u> -a                           | pirámid-e             |
| c. <i>athematic stem</i> | -cit-o/a                    | cafe- <u>cit</u> -o                             | café                  |
| $\neg n, \neg l, \neg r$ | -it-o/a                     | robot- <u>it</u> -o<br>virus- <u>it</u> -o      | robot<br>virus        |

### 3 STRATAL ANALYSIS

In Stratal Phonology, phonology applies iteratively over increasingly larger domains that derive from morphosyntactic structure in conformity with the general, presumably universal principles stated in (4). Each stratum may be associated with a distinct constraint hierarchy.

- (4) General procedure to define phonological domains (Bermúdez-Otero & Torres-Tamarit 2026)
- a. Every minimal constituent specified for a syntactic category (stem or affix) defines a cyclic domain for the stem-level phonology.
  - b. Every combination of a stem with a stem-level affix defines a cyclic domain for the stem-level phonology.
  - c. Every grammatical word (fully inflected lexical item) defines a cyclic domain for the word-level phonology.
  - d. Every utterance defines a cyclic domain for the phrase-level phonology.

According to these principles, each affix goes through its own stem-level phonological cycle. It will be demonstrated below that the diminutive affix in Spanish is a word-level affix; thus, allomorph selection of the diminutive takes place in the word-level phonology, once the stem and the affix are phonologically visible to each other. All tableaux below consist of word-level evaluations, and both the stem and the affix are fully syllabified forms, as indicated by the dots, because they have undergone separate stem-level evaluations. I use orthographic transcription throughout, and mark stress with an acute accent irrespective of orthographic rules.

As many others (see Smith 2011:§2), I propose that allomorph selection of the diminutive suffix is guided by a morphology-prosody alignment constraint requiring the right edge of the stem to align with the right edge of a syllable (ALIGN-R(stem, $\sigma$ ), Prince & Smolensky 1993/2004:127). Therefore, the *-cit-o/a* allomorph is selected by ordinary *e*-stems, which end in a consonant in the singular. This alignment constraint is defined in (5), and the evaluation of a diminutive derived from an ordinary *e*-stem is illustrated in (6). In the presence of two input stresses, only the last one survives (see Bermúdez-Otero & Torres-Tamarit 2026 for a comprehensive analysis of Spanish stress assignment within Stratal Phonology).



- (5)  $ALIGN\text{-}R(stem, \sigma)$  (based on  $ALIGN(stem, R; \omega, R)$  in Bermúdez-Otero 2012:46)

If  $o$  is the rightmost output segment standing in correspondence with some input segment  $i$  in a stem, then assign a violation mark if  $o$  is not rightmost in a syllable.

- (6) *cancioncita* ‘song.DIM’

|                                 |                                                                                                  |                           |
|---------------------------------|--------------------------------------------------------------------------------------------------|---------------------------|
| can.ción- $\{í.ta \sim cí.ta\}$ |                                                                                                  | $AL\text{-}R(st, \sigma)$ |
| a.                              |  can.cion.cí.ta |                           |
| b.                              | can.cio.ní.ta                                                                                    | *!                        |

In *e*-only stems, selecting either of the two allomorphs would satisfy the constraint  $ALIGN\text{-}R(stem, \sigma)$  equally well. However, another alignment constraint, formulated in (7), favors *-cit-o/a* over *-it-o/a*. This constraint requires vowel-initial, stem-based suffixes to align to the left with an onset, and is therefore responsible for stem-final vowel deletion in Spanish, an exceptionless process (e.g., *tax-ero* ‘taxi driver’, not *\*taxi-ero*, see Bermúdez-Otero 2013:47 for an extensive discussion).  $ALIGN\text{:}_{stem} \llbracket_{suffix} V \rightarrow_{onset} \rrbracket$  cannot be replaced by *\*HIATUS*, since hiatuses are permitted in Spanish at the lexical strata; stem-final vowel deletion is discussed extensively in Bermúdez-Otero (2013). The evaluation of a diminutive derived from an only *e*-stem is illustrated in (8).

- (7)  $ALIGN\text{:}_{stem} \llbracket_{suffix} V \rightarrow_{onset} \rrbracket$  (Bermúdez-Otero 2013:46)

For every vocoid  $V$  in the output, such that  $V$  has an input correspondent that is initial in a stem-attached suffix, assign one violation mark if the left edge of  $V$  does not coincide with the right edge of an onset.

(8) *fasecita* ‘phase.DIM’

| fá.se- $\{\acute{i}.ta \sim \acute{c}i.ta\}$ |                                                                                               | AL: $_{st} \llbracket_{suff} V \rightarrow_{ons} \rrbracket$ | AL-R(st, $\sigma$ ) |
|----------------------------------------------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------|---------------------|
| a.                                           |  fá.se.cí.ta |                                                              |                     |
| b.                                           | fá.sí.ta                                                                                      |                                                              | *!                  |
| c.                                           | fá.se.í.ta                                                                                    | *!                                                           |                     |

As opposed to *e*-only stems, *o/a*-stems select the *-it-o/a* allomorph. I interpret this as a strategy to avoid a certain degree of identity between the final stem theme vowel and the final affix theme vowel, according to a constraint I label OCP-ENDS (9). This constraint prohibits a syntagmatic configuration and should not be confused with antihomophony effects, as homophony is an intrinsically transderivational or paradigmatic relation. Such transparadigmatic relations are prohibited in Stratal Phonology, in production. The evaluation of a diminutive derived from an *o*-stem is illustrated in (10).

## (9) OCP-ENDS

Let *x* be an output segment in correspondence with a piece-final input segment *X*, and *y* an output segment in correspondence with a piece-final input segment *Y*. Assign one violation mark if *x* and *y* are both [+back, –high].<sup>2</sup>

 (10) ‘book.DIM’ *librito*

| lí.bro- $\{\acute{i}.to \sim \acute{c}i.to\}$ |                                                                                               | OCP-ENDS | AL-R(st, $\sigma$ ) |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------|----------|---------------------|
| a.                                            |  lí.brí.to |          | *                   |
| b.                                            | lí.bro.cí.to                                                                                  | *!       |                     |

We have seen that consonant-final, ordinary *e*-stems select *-cit-o/a* (6). However, those

<sup>2</sup>Diminutives such as \**manocita* ‘hand.DIM’ are also prohibited because both /o/ and /a/ are [+back] in Spanish. A diminutive derived from an athematic stem like *tribucita*, however, is allowed because /u/ is [+high].

that are monosyllabic select the same allomorph but surface with the theme vowel *-e*, which outside the diminutive appears only in the plural (1). I attribute this to faithfulness to the underlying prosodic specification of *-cit-o/a*, which is preceded by an empty binary foot in its underlying representation:  $/(\sigma\sigma)_{Ft}cit-o/$ . The need for lexical prosodic prespecification is independently motivated, as Spanish, for instance, has preaccented suffixes—affixes that require stress to fall on the immediately preceding syllable. These suffixes are stored with their first vowel linked to the weak branch of a foot.

When two stem allomorphs compete (e.g., *flor* and *flor-e*), the domain for selection is the cyclic domain triggered by the first syntactic operation on the stem—here, attachment of the diminutive suffix—which is associated with word-level phonology, as will be demonstrated. The candidate with the theme vowel *-e* and the *-cit-o/a* allomorph harmonically bounds all other candidates. The evaluation of a diminutive derived from a monosyllabic ordinary *e*-stem is illustrated in (11).

(11) *florecita* ‘flower.DIM’

|                                                   |                        | $ID-(\sigma\sigma)_{Ft}$<br>$AL:st \ll_{suff} V \rightarrow_{ons}$<br>$AL-R(st,\sigma)$ |    |    |
|---------------------------------------------------|------------------------|-----------------------------------------------------------------------------------------|----|----|
| {flór~fló.re}~{í.ta~ $(\sigma\sigma)_{Ft}cí.ta$ } |                        |                                                                                         |    |    |
| a.                                                | flór → flo.rí.ta       |                                                                                         |    | *! |
| b.                                                | flór → flor.cí.ta      | *!                                                                                      |    |    |
| c.                                                | fló.re → (flo.re)í.ta  |                                                                                         | *! |    |
| d.                                                | fló.re → (flo.re)cí.ta |                                                                                         |    |    |

Returning to *e*-only stems, which select *-cit-o/a* and surface with the theme vowel *-e*, those that are longer than two syllables instead select *-it-o/a* and surface without the theme vowel *-e*. I attribute this outcome to the satisfaction of the markedness rhythmic constraint \*LONGLAPSE, which disfavors unstressed syllables that are not adjacent to a

stressed syllable or to a word edge (Kager 2007).

(12) *piramidita* ‘pyramid.DIM’

|    | pi.rá.mi.de- $\{\acute{i}.ta \sim \acute{c}i.ta\}$                                               | *LONGLAPSE | AL-R(st, $\sigma$ ) |
|----|--------------------------------------------------------------------------------------------------|------------|---------------------|
| a. |  pi.ra.mi.dí.ta | *          | *                   |
| b. | pi.ra.mi.de.cí.ta                                                                                | **!        |                     |

Finally, *o/a*-stems containing a diphthong select *-cit-o/a*, rather than the default *-it-o/a*, and also surface with an [e] vowel, which I interpret as epenthetic (a DEP-V-violation), as opposed to the [e] vowels that appear in *e*-stems, which are thematic. On the one hand, the ending *-ecit-o/a* allows diphthongs to occupy a foot-head position (e.g., (*pie.dr-e*)(*cí.t-a*)), satisfying DIPHTHONG-HEAD (13), and on the other hand, to remain faithful to the non-syllabicity of the glide in the immediate base, satisfying IDENT[j] (16). The base with the diphthong is part of the input to the word-level phonology (e.g., (*mo.m[j]e*)(*cí.t-a*)  $\leftarrow$  *mó.m[j]a*). My analysis assumes iterative foot parsing, although only the rightmost foot survives; Spanish has only one stress per word (except syntagmatic compounds and adverbs ending in *-mente*). Notice that in the presence of an epenthetic vowel, the constraint ALIGN-Right(stem, $\sigma$ ) is violated.

(13) *momiecita* ‘mummy.DIM’

|    | móm[j <sub>1</sub> ]a- $\{\acute{i}_2ta \sim \acute{c}i_2ta\}$                                                                  | IDENT-[j] | DEP-V |
|----|---------------------------------------------------------------------------------------------------------------------------------|-----------|-------|
| a. |  mo.m[j <sub>1</sub> ]-e-cí <sub>2</sub> .ta |           | *     |
| b. | mo.m[í <sub>1,2</sub> ].ta                                                                                                      | *!        |       |

(14) *piedrecita* ‘stone.DIM’

|    |                                                                                                      | OCP-ENDS | DIPH-Hd | AL-R(st,σ) | DEP-V |
|----|------------------------------------------------------------------------------------------------------|----------|---------|------------|-------|
|    | p[j]édra- $\{\acute{\text{ita}}\sim\acute{\text{cíta}}\}$                                            |          |         |            |       |
| a. |  (p[j]e.dre)(cí.ta) |          |         | *          | *     |
| b. | p[j]e(drí.ta)                                                                                        |          | *!      | *          |       |
| c. | (pie.dra)(cí.ta)                                                                                     | *!       |         |            |       |

That the diminutive suffix is a word-level affix in Spanish is clear from the overapplication in the selection of the diphthongal stem in diminutives, as in *piedr-ecit-a*. Stems like this have two listed allomorphs:  $\{/\text{piedr-a}\sim\text{pedr-a}/\}$ . In stem-level derivation, the allomorph with the monophthong is selected because monophthongs are favored in unstressed position. However, in evaluative suffixation, which is word-level, the selection of the allomorph with diphthong “overapplies”, as illustrated in (15).

(15) Diphthongal stem alternations (Bermúdez-Otero 2013:§3.2)

| Stem-level        |                   | Word-level          |             |
|-------------------|-------------------|---------------------|-------------|
| <i>pedr-ad-a</i>  | ‘stone thrown’    | <i>piedr-ecit-a</i> | ‘stone.DIM’ |
| <i>a-pedr-ear</i> | ‘to stone’        | <i>piedr-az-a</i>   | ‘stone.AUG’ |
| <i>pedr-usc-o</i> | ‘unpolished rock’ | <i>piedr-ot-a</i>   | ‘stone.PEJ’ |

Stems that participate in the diphthongal alternation are special: their domain of selection is the stem-level phonology, even when there is no stem-level affixation, in which case allomorphy is resolved in a second cycle of the stem-level phonology (see Bermúdez-Otero 2013:§3.3 for a detailed discussion). The cyclic domain structure and the phonological derivation of *piedrecita*, which exhibits “overapplication”, are illustrated in (16).

(16) Cyclic domain structure (CDS) and phonological derivation of *piedrecita*

- a. CDS:  $\llbracket_{WL} \llbracket_{SL} \left\{ \llbracket_{SL} \text{pedr-a} \rrbracket \llbracket_{SL} \text{piedr-a} \rrbracket \right\} \rrbracket \left\{ \llbracket_{SL} \text{it-a} \rrbracket \llbracket_{SL} \text{cit-a} \rrbracket \right\} \rrbracket$
- b. *first cycle* outputs (SL-P):  $\left\{ \text{pédra; piédra} \right\} \mid \left\{ \text{íta; cíta} \right\}$
- c. *second cycle* output (SL-P): *piédra*
- d. *third cycle* output (WL-P): *piedrecíta*

To conclude this section, let me briefly refer to diminutives derived from athematic stems. Athematic stems seem to select *-cit-o/a* in order to satisfy  $\text{ALIGN-R}(\text{stem}, \sigma)$ . This is evident in vowel-final athematic stems (e.g., *cafecito*) and in athematic stems ending in coronal consonants other than /s/ (e.g., *stickercito*). Otherwise, they select *-it-o/a* (e.g., *robot-it-o*). I attribute this to satisfaction of the constraint \*CODA/obstruent (based on McCarthy 2008:228), as well as to the avoidance of [s.θ] clusters (e.g., *virus-ito*; cf. \**virus-cit-o*), which are otherwise allowed in the language (e.g., *piscina* [pis.θí.na] ‘swimming pool’), but never in diminutives. The evaluation of a diminutive derived from an athematic stem ending in an illicit consonant is illustrated in (17).

(17) *robotito* ‘robot.DIM’

|                                                                                        | ro.bót- $\{\acute{\text{í.to}} \sim \text{cí.to}\}$ | *CODA/obstr | AL-R(st,σ) |
|----------------------------------------------------------------------------------------|-----------------------------------------------------|-------------|------------|
| a.  | ro.bo.tí.to                                         |             | *          |
| b.                                                                                     | ro.bot.cí.to                                        | *!          |            |

There are still open issues that I have not addressed in this short paper. Ordinary *e*-stems ending in *-d* usually avoid diminutives (e.g., ?*pared-it-a*, \**pared-cit-a* ‘wall.DIM’), and those ending in /-l/ idiosyncratically select *-it-o/a* (e.g., *cartel-it-o* ‘poster/sign.DIM’, *hotel-it-o* ‘hotel.DIM’, *papel-it-o* ‘paper.DIM’; cf. *sal-e-cit-a* ‘salt.DIM’, derived from the monosyllabic ordinary *e*-stem *sal* ‘salt’).

## 4 CONCLUSIONS

Unlike many previous constraint-based, monostratal approaches (see Smith 2011:§2), this analysis of diminutive formation in Peninsular Spanish upholds a high standard of modularity.

First, no morphological (e.g., nominal class) or syntactic property (e.g., DIM), is named in the statement of the phonological constraints, except for the broad distinction between stems and affixes in alignment constraints (in compliance with the Indirect Reference Hypothesis, Bermúdez-Otero 2012:77).

Secondly, the high standard of modularity is also grounded in cyclicity: morphosyntactic structure determines the cyclic domains in which allomorphy is resolved by controlling the size and stratal affiliation of domains for phonological computation.

The analysis also admits lexical prespecification for the allomorph *-cit-o/a*, which needs to be preceded by at least a disyllabic foot. Lexical prosodic prespecification is a principled mechanism for accounting for phonological nonuniformity.

In Stratal Phonology, the phonology only sees the boundaries between input pieces and distinguishes between stems (projections containing non-functional, root material) and non-stems, or affixes, by only referring to the predicate ‘stem’, and not morphosyntactic features or specific morphemes. In any event, it is widely accepted that phonology must make reference to the lexical-functional divide, since robust crosslinguistic data clearly demonstrate systematic phonological differences between stems and affixes.

The formulation of the alignment constraints also respects inward cyclicity because it only makes reference to the morphological boundary of the stem, not between more internal boundaries within those pieces, like those between roots and theme vowels.

Phonology must likewise differentiate structures that are internal to a single input unit from those created through the combination of multiple input units, whose size may differ across strata. Within the framework of the stratal theory of morphological

colors (Trommer 2015), this contrast can be formally captured in a constrained way by appealing to the predicates ‘tautochromatic’ and ‘heterochromatic’. The constraint OCP-ENDS is consistent with this, as it only refers to pieces of different color.

Hopefully, I have demonstrated that an articulated theory of nominal classes in Spanish and a correct understanding of cyclic domain structure (Bermúdez-Otero 2013) allow for a comprehensive account of diminutive formation in Peninsular Spanish that upholds a high standard of modularity, as opposed to previous constraint-based accounts.

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